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FREE LIVE SESSION · 30 MINUTES

# FinOps for AI-ML Systems

Read an AI cloud bill like an owner, cut the waste without hurting the model, and keep the savings.

**PRESENTER** Dr. Emmanuel Naweji **LIVE** Mondays & Thursdays · 7:30 PM Central · 30 min

TONIGHT'S 30 MINUTES

# Welcome! Here's the plan

## 0–2 min

### Welcome

Say hi in the chat: your name, where you're joining from, and your role.

## 2–20 min

### Three modules

One real idea from each module of FinOps for AI-ML Systems, with the lab behind it.

## 20–23 min

### Keep going

How to get the full course, every lab, and support along the way.

## 23–30 min

### Your questions

Live Q&A. Drop questions in the chat any time; I'll answer at the end.

WHY THIS COURSE MATTERS

# AI bills leak, a little every hour, from things nobody owns

01

## GPUs bill by the hour

Busy or idle, the meter runs. A forgotten notebook GPU costs as much as a busy one.

02

## LLMs bill by the token

Every word in and out has a price. Usage grows quietly with success.

03

## Experiments sprawl

Notebooks, snapshots, test endpoints: created fast, deleted never, tagged rarely.

THE COURSE MAP

# Inform, Optimize, Operate: three modules

MODULE 1

## See Every Dollar

Read a month of AI bills: where the money goes, who owns it, what one prediction costs.

MODULE 2

## Cut the Waste, Not the Model

Stop idle GPUs, train on spot safely, and cut the LLM bill above a quality floor.

MODULE 3

## Budgets That Bite

Forecast every team's month, catch spikes on day one, and gate new spend before it exists.

01

MODULE 1 OF 3

# See Every Dollar

Read an AI cloud bill like an owner. You can't cut what you can't see.

# Where this month's $5,778 went

**$2,160**

gpu.small: the notebook GPUs, 37% of the bill

**$1,384**

gpu.large: actually training the model

**$1,214**

llm.large: the AI assistant, paid per token

**28%**

of all spend has no owner tag at all

The notebooks cost more than training the model. Nobody decided that. Nobody was looking.

# Three tags, on everything, from day one

01

## team

Who pays and who gets the alert: fraud, research, support.

02

## model

Which AI system it serves: fraud-v2, assistant. Needed for unit costs.

03

## env

dev or prod. Different rules: dev GPUs must stop themselves.

Target: at most 5% of spend untagged. Our lab starts at 28%. No tag = no bill, no alert, nobody turns it off.

# Your mission: read the bill like an owner

1

## Where does the money go?

top.py: the total, cost by SKU, and the top 5 resources.

2

## Who owns this?

tag_audit.py: untagged share. PASS at 5% or less, else exit 1.

3

## Fix tags, then showback

Apply tag fixes, then show each team its own bill. UNALLOCATED stays visible.

4

## What does one unit cost?

unit_cost.py: cost per 1,000 predictions and per chat.

# Give every dollar an owner

TERMINAL

```
$ python3 lab/make_data.py
$ python3 module-1/tag_audit.py; echo "exit $?"
$ cp module-1/tags.example.json module-1/tags.json
$ python3 module-1/showback.py
```

**`make_data.py`**
a month of pretend bills; same seed, same numbers

**`tag_audit.py`**
28% untagged: RESULT FAIL, exit 1

**`tags.json`**
fills in missing tags only

**`showback.py`**
each team's bill, UNALLOCATED kept visible

YOU SHOULD SEE RESULT: PASS after the fix, and research as the biggest spender at $2,273, more than the production fraud team.

# Three numbers your CFO will understand

**$0.077**

per 1,000 fraud predictions, to serve

**$0.270**

per 1,000 predictions, including training

**$0.012**

per assistant chat

Total cost rises with success. Unit cost shows whether you're getting more efficient, or just bigger.



# You can't cut what you can't see.

Where, who, and per what. Every month. Then ask of every new resource: who will turn this off?

02

MODULE 2 OF 3

# Cut the Waste, Not the Model

Idle GPUs, spot training, cheaper LLM calls: with a quality floor.

# Waste has three shapes: look in this order

01

## Idle

Paying for capacity that does nothing: notebooks overnight, forgotten GPUs. Easiest win.

02

## Overpriced

Paying full price for work that could run on cheaper, interruptible capacity.

03

## Oversized

A huge model, or a huge machine, for a small job.

In the lab, idle.py finds ≈ $1,897 a month: two notebooks to AUTO-STOP, one to DELETE.

# Spot is cheap only if you can survive losing it

### ON-DEMAND

**$4.00 / hour**

Yours until you stop it. Full price. Right for production endpoints. In the sim: 40 h, $160.

VS

### SPOT

**$1.20 / hour**

Spare capacity that can be taken back. No checkpoints: $275, 24% never finish. Hourly checkpoints: $52.

Lab prices, 500 simulated runs each. Same discount, opposite results: spot savings come from your code, not the price list.

# Cheaper LLM bills, same quality floor

**$1,214**

all large model: 97.9% good answers · PASS

**$121**

all small model: 90.2% good answers · FAIL

**$290**

route easy questions to small: 97.2% · PASS

**$102**

route + cache good answers: 98.9% · PASS

Floor: at least 95% good answers. Route + cache is 92% cheaper with better quality. Never cache anything personal.



# “What's the worst quality we'll accept?”

Agree on it before you look at the savings. A saving that breaks the floor isn't a saving: it's a cost moved onto your customers.

# Your mission: find $1,897 and cut the LLM bill

1

## Find the idle GPUs

idle.py: busy vs idle hours per GPU, DELETE / AUTO-STOP / ok, and the saving.

2

## Is spot worth it?

spot_sim.py: on-demand vs spot vs spot + checkpoints, 500 runs each.

3

## Cut the LLM bill

llm_router.py: four strategies on 2,000 requests, against a 95% floor.

# Pay for work, not for waiting

TERMINAL

```
$ python3 module-2/idle.py
$ python3 module-2/spot_sim.py
$ python3 module-2/llm_router.py
```

**idle.py**
an hour is busy at 10%+ GPU use

**spot_sim.py**
lost work + restart time, billed on the clock

**llm_router.py**
verdict = PASS only if quality >= 0.95

YOU SHOULD SEE notebook-gpu-03 DELETE · Estimated saving: $1,897 per month · route + cache $102 98.9% PASS

03

MODULE 3 OF 3

# Budgets That Bite

Forecasts, alerts and a cost gate that keep the savings.

# A clean-up vs a system

A CLEAN-UP

## Once a quarter

Someone finds the waste, deletes it, sends a happy email. New waste starts the next day. Savings fade.

VS

A SYSTEM

## Every day, automatically

Forecasts, alerts to owners, and a gate on new resources. Waste is caught as it starts. Savings stick.

# Day 20 of 30: who will blow the budget?

**374%**

research: forecast $2,988 vs an $800 budget · OVER

**96%**

support: the assistant keeps growing · WARN

**86%**

fraud: on track · OK

**36x**

research spike on day 74: $791 vs a normal $22

Forecast = spent so far + median of the last 7 days × days left. Spike = over 1.5× normal AND at least $50 more.

# Three rules, checked before anything is created

01 **Every resource tagged**
team, model, env, or FAIL. Module 1, enforced.

02 **Dev GPU → auto_stop**
No always-on dev GPUs. Module 2, enforced.

03 **Fits the budget**
Forecast + new cost ≤ budget, else owner approval. Not 'no': 'ask first'.

04 **change-ok.json**
A 4th fraud-api server: +$144 a month. Tagged, fits. PASS, exit 0.

05 **change-bad.json**
4 big always-on GPU notebooks: +$11,520 a month. FAIL ×3, exit 1.

The cheapest dollar is the one you never start spending.

# Forecast, alert, gate

TERMINAL

```
$ python3 module-3/forecast.py
$ python3 module-3/anomaly.py
$ python3 module-3/cost_gate.py module-3/change-bad.json
$ python3 -m unittest discover tests
```

**forecast.py**
OK / WARN / OVER / NO OWNER per team

**anomaly.py**
spike → alert to the owner in budgets.json

**cost_gate.py**
tags, auto_stop, budget headroom; exit 0/1

**unittest**
6 robot checks on the guardrails

YOU SHOULD SEE ALERT day 74: research spent $791 (normal $22, 36x) · change-bad: +$11,520 per month, FAIL ×3 · Ran 6 tests OK

# Daily, weekly, monthly, on every change

1

## Daily: anomalies

anomaly.py → alerts to owners. Fixed the same day.

2

## Weekly: showback

Each team sees its bill and unit cost. 15 minutes.

3

## Monthly: forecast + budget

WARN/OVER teams decide: cut, or raise the budget with a reason.

4

## Every change: the gate

cost_gate.py in CI. Tags and budgets checked before anything exists.



# Every dollar needs an owner and an alarm.

Forecast it, alert on it, and gate it, so the savings stay saved.

RECAP

# The whole course on one slide

## You can't cut what you can't see

Module 1: where, who, per what. Three tags, showback, unit cost.

## Pay for work, not for waiting

Module 2: idle GPUs, spot with checkpoints, route and reuse, quality floor.

## Every dollar: an owner and an alarm

Module 3: forecast, alert, gate, and a weekly rhythm.



# Find its owner, its unit cost, and its alarm.

Pick one system you touch this week. Whatever's missing is your first FinOps win, and a great story for your next interview.

THE MEMBERSHIP

# Every course. One price.

**$49**

**per month**

All 8 courses, every module and every lab, plus a live session with me every month.

ALL 8 COURSES INCLUDED

- Linux for AI-ML Systems
- Python for AI-ML
- Zero to AI-ML Systems Engineering
- DevOps for AI-ML Systems
- SRE for AI-ML Systems
- GPU Optimization for AI-ML Systems
- FinOps for AI-ML Systems
- AI Agent Operations

BUILD WITH ME, LIVE

# Once a month, we build something together

1

## Learn at your pace

Work through any course, any module, on your own schedule.

2

## Bring a real problem

An idea, a project at work, or a lab you want to take further.

3

## Meet me live

Once a month we get on a call and build it together, step by step.

4

## Leave with it built

Something real you made, understand and can show.

FREE LIVE SESSIONS

# Every Monday and Thursday, 7:30 PM Central

A different course each session. The full rotation takes 4 weeks, then starts again.

| | WEEK 1 | WEEK 2 | WEEK 3 | WEEK 4 |
|---|---|---|---|---|
| MONDAY | Linux for AI-ML Systems | Zero to AI-ML Systems Engineering | SRE for AI-ML Systems | TONIGHT<br>FinOps for AI-ML Systems |
| THURSDAY | Python for AI-ML | DevOps for AI-ML Systems | GPU Optimization for AI-ML Systems | NEXT<br>AI Agent Operations |

# Join the membership today

**$49/month** · all 8 courses · a live build session with me every month

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Next free session: Thursday, 7:30 PM Central · AI Agent Operations

**Questions? Let's take them now.**

Dr. Emmanuel Naweji · Transformed 2 Succeed